

FELIPE SANTOS

Physical AI | Edge AI | AIoT Platforms | Solutions Architect • Tech Lead • Practice Lead | M.Sc. in AIoT & Cybersecurity

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EXECUTIVE SUMMARY

Senior technical leader with 7+ years architecting intelligent systems that connect the physical and digital worlds — Physical AI, Edge AI, AIoT, computer vision, robotics, and agentic AI platforms with LLM reasoning cores. Practice Lead for Edge AI & Platforms within a 50+ person Physical AI Studio at Globant and member of the studio management team. Proven record leading cross-functional teams of up to 8 engineers across AWS, Azure, and GCP, delivering mission-critical platforms for global brands (LATAM Airlines, NBA/LA Clippers, Universal, Disney). Trusted client-facing solutions architect driving presales wins, PoCs, MVPs and multi-project account growth. AI-native engineering practitioner: agentic coding workflows, spec-driven development, context engineering, and AI agent harness design for record-speed delivery. Mentor to multiple engineers, technical hiring gatekeeper (10+ interviews), and multilingual communicator (EN/PT/ES).

CORE COMPETENCIES

AI & ML: Physical AI • Edge AI • Agentic AI & Autonomous Systems • LLM Reasoning & Multi-Agent Orchestration • GenAI Platform Architecture • Computer Vision (YOLO, OpenCV, NVIDIA DeepStream, TensorRT, Triton, OpenVINO, custom model training) • MLOps (SageMaker, edge-to-cloud pipelines) • PyTorch • TensorFlow • Sensor Fusion • Reinforcement Learning • Real-Time Inference Optimization

IoT & Edge: AIoT Platforms • IoT Device Management at Scale (20,000+ devices) • Azure IoT Hub/Edge • AWS IoT Core • OTA Updates (Balena, Mender) • Zero-Touch Provisioning • MQTT/Mosquitto • BLE Beacons & Gateways • NVIDIA Jetson • Raspberry Pi • Embedded Linux • Robotics • Digital Twins • Industrial Edge Computing • IoT Cybersecurity

Cloud & Platform: Multi-Cloud (AWS, Azure, GCP) • CloudOps • Terraform/IaC • Docker • Kubernetes • CI/CD • Microservices & Event-Driven Architecture • Serverless (Cloud Run, Lambda, ECS) • High-Throughput Data Ingestion (Pub/Sub, Dataflow, Kinesis Firehose) • Go • Python • Bash • SQL/NoSQL (Firestore, DynamoDB)

Leadership & AI-Native Delivery: Solutions Architecture • Presales & Technical Storytelling • Technical Roadmaps • Team Leadership • Stakeholder Management • Executive Demos • Vendor Management • Staffing Strategy • Mentoring • Technical Hiring • Agentic Coding (Claude Code, Codex, OpenCode) • Spec-Driven Development • Context & Token Engineering • Agent Harness Design

PROFESSIONAL EXPERIENCE

GLOBALANT — Tech Lead • Solutions Architect • Practice Lead

May 2022 – Present

Physical AI Studio (official title: Technology Consultant) — Edge AI, AIoT, computer vision, and robotics solutions for aviation, sports & entertainment, smart venues, utilities, and manufacturing.

Studio & Practice Leadership

- Practice Lead for Edge AI and Platforms verticals within a 50+ person Physical AI Studio; contributor to Robotics and Embedded practices.
- Member of the studio management team: drive hackathons, tech sessions, internal R&D prototypes, and staffing strategy for studio projects.
- Tech Lead of internal product initiatives: RobOps platform and intelligent edge devices for human-robot interaction (Edge AI + robotics prototypes).
- Solutions Architect in presales: technical pitches, solution design, PoCs, MVPs, and live demos contributing to multiple six-figure USD project wins, plus account extensions and upsells.
- Mentor to multiple engineers and studio hiring gatekeeper: 10+ technical interviews for staffing candidates.

LATAM Airlines — Lead Architect & Engineer, Agentic AI Operations Platform

2026 – Present

- Principal architect of an AI platform with an LLM reasoning core for real-time monitoring, management, and control of airport turnaround operations — anomaly detection, operational risk identification, and autonomous decision-making.
- Designed event-driven GCP architecture (Compute Engine, Pub/Sub, Cloud Run APIs, Dataflow, Firestore) in Go and Python with multi-source sensor fusion: computer vision vendors, airline subsystems, and a proprietary BLE beacon/gateway solution for people and asset tracking.

Universal Parks — Jurassic Park Ride | Tech Lead, Architect & Principal Engineer

2025 – 2026

- Architected a real-time computer vision safety and operational-efficiency platform at the edge: left-object detection, people counting, restricted-zone monitoring, and ride dispatch metrics, integrated with operator alert panels.
- Achieved 0.95+ F1-score with <500 ms end-to-end latency using a custom-trained YOLO model on NVIDIA DeepStream with optimized edge inference (industrial PCs, MQTT/Mosquitto, Go, AWS, Docker, CI/CD); led a team of 5 engineers with on-site demos and client-facing delivery.

BBVA Argentina — Innovation Consultant (Internal Hackathon)

2025

- Advised the team on an eye-tracking applied-intelligence and cybersecurity solution for the bank's mobile application.

Compass (Cosan) — Tech Lead, Smart Metering AIoT Platform

2024 – 2025

- Led a team of 5 building an IoT platform to integrate and remotely manage millions of smart meters: automated device provisioning, device management, security, and OTA updates.
- Engineered high-throughput IoT data ingestion and preprocessing on AWS (IoT Core, Kinesis Firehose, Lambda, ECS, DynamoDB, S3, MQTT) with Terraform IaC in Go/Python; integrated with the data organization for advanced analytics and data intelligence.

LA Clippers (NBA) — Intuit Dome | Senior Engineer promoted to Tech Lead

2023 – 2024

- Built and led delivery of an enterprise device management platform for 20,000+ devices at the Intuit Dome smart venue — remote control, fleet monitoring, OTA updates, zero-touch provisioning, cybersecurity, and automated deployments.
- Led the IoT Edge and IoT Platform engineering teams on the full Azure IoT ecosystem (IoT Hub, IoT Edge) with Terraform, Docker, Python, and CloudOps; promoted from engineer to tech lead during the engagement; on-site delivery and client strategy participation.

HME — Edge-to-Cloud MLOps Platform Engineer

2023

- Developed an end-to-end edge-to-cloud MLOps platform for computer vision models on the NVIDIA stack (DeepStream, YOLO) with Python, Docker, and Bash.

Disney Parks — MLOps Platform Engineer

2022 – 2023

- Built the end-to-end MLOps platform (deployment, monitoring, data consumption, training, and inference) for a computer vision solution improving ride operational efficiency — AWS SageMaker, DynamoDB, Terraform, Python, CI/CD.

BRPHOTONICS — Automation Analyst promoted to Automation & Control Engineer

Apr 2020 – Apr 2022

R&D and manufacturing of optoelectronic devices (lasers, amplifiers) for optical communications.

- Designed and fully automated manufacturing, testing, calibration, and quality processes for lasers, amplifiers, and other optoelectronic devices; architected pilot production lines from R&D approval to manufacturing scale.
- Led junior engineers, interns, and technicians across automation, test, and quality engineering; time-series and statistical data analysis for product performance validation.

IDEA! ELECTRONIC SYSTEMS — Product Engineering Intern

Jul 2019 – Mar 2020

- Automated calibration, testing, and quality processes for 5G infrastructure optoelectronic devices.

EDUCATION & RESEARCH

M.Sc. in Computer Engineering — UNICAMP (University of Campinas), Brazil

2024 – 2026

- Thesis: Reinforcement Learning applied to cybersecurity in IoT device networks — designed and built an RL agent training framework for autonomous cyber-attack response using the military Kill Chain taxonomy.
- Paper under review at Elsevier *Internet of Things*: [view paper \(SSRN\)](#)
- Collaborated on graduate research projects in digital twins, advanced data analytics, robotics, and AI.

B.Sc. in Electrical Engineering (Control & Automation) — UFPB, Brazil

2014 – 2020

- CNPq/PIBIC research fellow (2016–2019): ultrasonic anemometry for wind velocity measurement; member of the Applied Computational Intelligence group; 4 peer-reviewed publications (IEEE, IOP): [IEEE \(2018\)](#) · [IEEE \(2019\)](#) · [IOP J. Phys. \(2018\)](#)
- Science Without Borders scholar (CAPES): exchange year in Electrical & Computer Engineering at the University of Wisconsin – Milwaukee, USA (2015–2016); Assistant Researcher at Mercer University, USA (2016) — wireless power transfer via resonant inverters.

Languages: Portuguese (native) · English (fluent, C2) · Spanish (fluent, C2)